

new LoRA adapter served to vLLM (generation and training share one GPU)

Batch

$B = 12$ cases drawn by weight: CDM 0.75, OTHER 0.15, controls 0.10

Rollouts

$G = 8$ multi-turn episodes per case, sampled by vLLM with the current adapter

Rewards

the commit reward; for each **DEFER**, $K = 3$ forced continuations from its state

Advantages

$r - \bar{r}_{\text{group}}$ on every turn; the **DEFER** turn takes $A = E - \hat{V}(s)$

Policy update

clipped token-level surrogate; truncated importance weights against the engine

measure

In-set deferral rate

ρ_k : share of the batch's in-set episodes that end in **DEFER**

PI coverage controller (Stooke et al., 2020)

$$e_k = \rho_k - \rho_{\max}$$

$$l_k = \text{clip}(l_{k-1} + e_k, 0, \nu_{\max}/K_i)$$

$$\nu_k = \text{clip}(K_p e_k + K_i l_k, 0, \nu_{\max})$$

$$\rho_{\max} = 0.30, K_p = 1.5, K_i = 0.2, \nu_{\max} = 2$$

ν_k enters E

ν_k is charged to **DEFER** on in-set cases only; deferring an OTHER case is never rationed.

The comparator arm uses plain dual ascent:
 $\nu \leftarrow \max(0, \nu + 2 e_k)$.